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Smart Homes

Human interactions and IoT

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Abstract

This thesis studies the phenomenon of human-system interaction in smart homes as a practice of Internet of Things. The research was conducted through interviews, workshops and observations and followed an interpretive research paradigm of phenomenologically-situated paradigm of HCI and a qualitative research approach. The theories of Phenomenology and Postphenomenology were used to interpret the experiences, beliefs and views of the participants. The empirical findings were processed and a thematic analysis was followed in order to identify the main themes that emerged out of the interviews, workshops and observations. The discussion of the findings showed that the research questions were answered to the grade that the participants of the research design, use and interact with smart homes in a multiple and complex way.

Keywords: Human-System Interaction, Smart Homes, Internet of Things, Postphenomenology, Technological Mediation

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1. Introduction

People and organizations use Information and Communications Technologies (ICT) in any aspect of their professional and private life. This extended use is possible through wireless communications e.g. smartphones and networks. The use of ICT produces social and technological implications, but at the same time offers a vision for the development of a good ICT society (Walsham, 2001). Bradley (2010) suggests a convergence model to illustrate how the design and use of ICT have evolved, which shows how human beings and society converge and interact. Convergences are present in a set of working and daily life activities. The use of information and communication technologies provides people with the capabilities to work with tasks in a different way (Bradley, 2010). For example the writing of a book or an architectural drawing is now a work supported by computers and further is disseminated via web or other communication technologies. Internet of things (IoT) breaks out in a continuous process due to the convergence of information and communication technologies.

There is not a wide accepted definition of Internet of Things but it could be described as Chen and Jin (2012) suggests:

"network which can achieve interconnection of all things anywhere, anytime with complete awareness, reliable transmission, accurate control, intelligent processing and other characteristics by the supportive technologies, such as micro-sensors, RFID, wireless sensor network technology, intelligent embedded technologies, Internet technologies, integrated intelligent processing technology, nanotechnology" (Chen and Jin, 2012, p. 562)

Another more open and flexible definition is that Internet of Things is the technological infrastructure that permits "people and things to be connected any-time, any-place, with anything and anyone, ideally using any path/network and any service" (Vermesan et al., 2009, p. 12). The development of IoT raises many considerations about the issues of privacy, security, networking and the use of international standards in order to achieve the needed level of interoperability (Atzori, Iera and Morabito, 2010, pp. 2787-2788).

Interconnectivity is a core issue in IoT infrastructure that has both theoretical and technological implications. Infrastructure could be defined as George Bateson (1978 cited in Star and Ruhleder, 1996, p. 112) suggests "What can be studied is always a relationship or an infinite regress of relationships. Never a «thing»". These relationships are developed within a wide scale of actions, plugs, build environment and bureaucratic forms (Star, 1999). In addition, the Information Infrastructure (II) is an integrated or merged entity between information and infrastructure technologies with some critical aspects which indicate that II is: enabling, shared and open, has heterogeneity and installed base (Hanseth and Monteiro, 1998). Regarding the IoT infrastructure the network is made by the use of some technologies that permit the

marking and identification along with the localization and the tracing of the physical objects. A wide spread technology for this implementation is the Radio Frequency Identification (RFID) system and the Wireless Sensor Network WSN (Zhang and Zhu, 2011). The net of sensors and "smart objects" constitute the lower level of infrastructure of IoT (sensing layer). The upper level is the corpus of applications that derives data from things and promotes services. In addition, there is a need for "communication" between these levels that is made real by the development of a middleware. Middleware is a software that is able to interoperate with the advanced and complex level of application and Operating Systems (OS) and the simple and heterogeneous hardware of things (Azzara et al., 2013). Middleware is intended to fill this lack of communication among the parties of IoT.

Another approach of IoT, suggested by some researchers, is the Social Internet of Things (SIoT). In SIoT the objects (smart devices) develop "relationships" with other objects and interchange data in order to resolve problems that their owners face (Atzori, Iera and Morabito, 2011). These interchanges are conducted when the objects have the permission by the owners. There are different types of objects "social relationships" like the *parental object relationship*, when the objects have been manufactured in the same factory and at the same period and the *co-work object relationship*, when the devices behave as co-workers in the same place. For example a *parental object relationship* helps, in a case of a service request, to exploit the "experience" of an item from the same manufacturer who has already addressed the problem. In addition, a *co-work object relationship* is developed with objects like sensors, RFID tags, actuators that constantly or periodically cooperate in the same place in order to provide an IoT application such as an emergency response or telemedicine (Atzori et al., 2012).

Smart homes are an appealing Internet of Things practice. The integration of IoT creates a home environment where the incorporated sensors and actuators are self-configured and can be accessed and controlled through Internet offering the users a variety of monitoring and control applications (Li et al., 2011). The collected data by the RFID "smart devices" are transmitted through the focal nodes to the web applications allowing the users to interact and control them (Welbourne et al., 2009; Qin and Gu, 2011). The net of sensors and RFIDs receives data and transfers them to a router and then to the IoT middleware. After that, the data are transferred to the IoT Domain Name System (DNS) and finally to the IoT IS. Then the system responds, with a correspondence message, to the users about the detail information of the objects.

Beyond the technological infrastructure of smart homes there are some other components that contribute in understanding the reasons that motivate the residents to use these facilities offered by IoT. There are different types of smart home services and functions like security, healthcare and energy saving implementations (De Silva, Morikawa and Petra, 2012). Smart homes support multiple human actions and a wide range of residents' needs, offering convenience, easy living, entertainment and home

control. From the researchers view “it appears that healthcare and wellbeing, as well as home utilities, are the most interest” in the process of smart homes ubiquitous and pervasive technologies environment (Koreshoff, Robertson and Leong, 2013, p. 341).

Furthermore, there are some factors that facilitate or hindrance the development of smart homes markets. The way of implementation is a key technical point that affects the adoption of smart homes. The decision of residents to upgrade an existing home or to obtain a purpose-built home from scratch affects the process of equipment installation and cost. Some other components that affect the decision to live in a smart home are the interoperability of infrastructure, the reliability of home functions, the privacy and security issues of personal data, and the cost of equipment and installation (Balta-Ozkan, Boteler and Amerighi, 2014). In addition, the above context is described more vividly in the view of the residents’ beliefs as these are formulated by Brush et al. (2011).

Smart homes are a practice of IoT that deploys issues for research. A significant topic of research is the HCI in smart homes. The design of smart homes affects the way humans interact with their home space and devices. Some researchers suggest four critical factors that should be considered in the design process of these systems: properties of users, requirements of services, the status of environment and the interaction level of a space (Wu and Fu, 2012). Other researchers indicate further investigation of smart homes design that focus on the services such as security, energy saving, entertainment and information. Furthermore, the human-computer interaction within this computerized home environment is a critical issue. The humans are surrounded by physical and artificial environments that form their acts and reacts on the environments’ conditions (Miorandi et al., 2012). Residents desire to exploit a set of services and functions that are offered in smart homes. The services that the system provides, and the way that services interact with the residents, are interdependent components in human system interaction in smart homes. To consider these aspects scientistist suggest a “living laboratory” to realize the assessment of human interaction with the system in order to make people learn from the technology and the services (Intille, 2006).

1.1The Aim and Research Questions

Many researches about smart homes focus on humans’ privacy, personal usage levels, ethical problems, multiple spaces, service priorities and legal implications (Weber, 2009; Wu and Fu, 2012). Nevertheless, IoT research is dominated by a technological focus, consequently there is a need to develop human-centred IoT (Koreshoff, Leong and Robertson, 2013). This is motivated by an underestimated focus on the human agency and how it is configured in these technologies. The human-centred interaction in the design of IoT-related devices emerges also in the review of literature as Koreshoff, Leong and Robertson (2013) indicate.

The aim of the study is to investigate Internet of Things as an emergent research area with a specific focus on the design and implementation of smart homes. This

research focuses on humans interactions in smart homes. Thus, it is intended to understand the way the designers and implementers design and install various services in the smart homes and how these technologies influence prospect residents' views of smart homes.

The research questions are:
In which way are designers and prospective residents involved in the design of smart homes?
What kind of services and human-system interactions do designers include into the design of smart homes?
What visions and ideas do prospective residents express about life in smart homes?

The study of humans interactions in smart homes has a great significance as Human Computer Interaction is a research topic that concerns both with understanding of how people make use of devices and systems that incorporate computation and with design of new devices and services.

1.2 Topic Justification

There is a discussion among researchers regarding the framework of human interactions between smart homes and residents in order to fulfill comfort, convenience and security. The proposed frameworks focus on the context of services, spaces, and users, and their interaction is analyzed (Wu and Fu, 2012). The tendency to broaden computing away from the workplace continues and new research issues arise. A research topic is the focus on humans needs in their everyday lives, specifically, that part of their lives' that doesn't constitute work. For this reason, there are many efforts in investigating research issues that focus on IoT in smart homes. Researchers study smart homes implementations that serve as a living laboratory for ubiquitous computing in the support of home life. These smart homes produce an environment that is capable of knowing information about itself and the whereabouts and activities of its residents (Kidd et al., 1999).

1.3 Scope and Limitations

In this study I decided to focus on the designers of smart homes, since they have a critical contribution in the implementation of services included in smart homes. The included services affect residents' interaction with these systems. In addition, my intention was to include smart home residents or prospective residents. However, in the process of this research I faced significant difficulties to gain access to the residents of smart homes due to the negative stance of the companies to provide me any contact information of their customers. So, having in mind that is important to have the residents' view of the life in smart homes, I decided to conduct workshops with ordinary people from my social and working environment to explore their views and ideas of smart homes.

The rest of the thesis is structured as follows. In Chapter 2, I present the Theoretical framework and the theories that I used. In Chapter 3, it is reviewed the literature about the topic of the thesis. Further, Chapter 4 is referred to the methodology of the research, the research setting, the research paradigm, the data collection methods and the data analysis. In Chapter 5, the empirical findings of this research are presented. In addition the findings of the research are discussed in Chapter 6. Finally, in Chapter 7 it is provided a conclusion of the research.

2. Theoretical framework

In this chapter are presented and discussed the theories that will be used.

I decided to use *phenomenology* and *postphenomenology* because they are two philosophies that deal with human experiences and technology as two inderdependent components. Below follows a presentation of the theories and the relationship between them. Also, through the presentation of the theories are explained the reasons that the use of these theories helps to investigate the topic of Human-System Interaction and the Internet of Things in relation to the humans' activities and experiences. Phenomenology and postphenomenology are presented first and then follows a presentation of philosophy of technology. Finally the chapter ends with a conclusion.

2.1 Phenomenology and Postphenomenology

Phenomenology is a philosophy and a method of investigation. It has its origin in the work of Husserl and was developed by Heidegger and Marleau-Ponty (Introna and Ilharco, 2004). For Edmund Husserl the meaning of objects is the understanding by our senses as a *flux of unordered sensations*. The meaning of the things in any particular time depends on the remainder, an already there *sense* that we have for the objects. Any experience of the world remains to our consciousness as *taken*, as assumption of a given object. This categorization of the things in our minds produces the *transcendental* domain, the sense of the ongoing experience that helps us to perceive the world as it is (Introna and Ilharco, 2004). Also, this philosophical theory introduces the perspective to investigate the essence of human experience and the terms of intentionality, meaning, intentional object, noema, noesis and beliefs. All the above components are useful to the research of human-system interaction. Husserl developed the Phenomenology introducing the term of *intentionality* of human conciousness. Further, in Husserl terms, the "phenomenological reduction requires that one suspend all forms of theorizing and generalization, even the supposition that something has to be somehow 'real' or 'concretely existing' to be experienced" (Introna and Ilharco, 2004, p. 66). This conception guides to the inference that in phenomenological reduction the subject of attention is not longer this or that actual thing, event or situation but the phenomenological *meaning* of these entities as they *show themselves* across all our intentional experiences.

Don Ihde is the founder of the theory of postphenomenology. In his work *Postphenomenology and Technoscience* (2009), Ihde makes a reappraisal of the movement of phenomenology across the 20th century, with a thorough comparison between phenomenology and pragmatism and he points out the key concepts that he indicates as a new perspective of the theory that calls postphenomenology. This theory suggests that the meaning of human practice and interpretation of the things is dominated by factors such as the embodiment of the materials/things, the technique

of use of them and the cultural-historical role of the environment where an *experience* is generated (Ihde, 2009, p. 18-19). Moreover, the postphenomenology intends to explore the impact of technology to the *lifeworld* of humans' experiences. Our understanding of the reality depends on the personal, cultural and social factors and the perception of body in relation to space and artifacts.

2.2 Philosophy of Technology

The world of humans has been bounded with the technology since the Titan Prometheus gave the fire to the humans, the forbidden secret of the Gods, as the Greek Mythology narrates. The fire was the primary element that permitted the humans to create the crafts. Technology has a long and consistent place in the philosophical thinking, from the ancient Greek philosophers Plato and Aristotle to the most prominent modern philosophers like Karl Marx, Martin Heidegger, Edmund Husserl and Michel Foucault (Franssen, Lokhorst and Van de Poel, 2009). For the ancient philosophers the phenomenon of technology was approached with variable forms and produced different meanings. For Plato the technology learns from the nature or imitates it. Also, Democritus saw the technology as an applied human practice that takes the physical environment as pattern. In his perspective the building of a house and the creation of textiles are imitations of the work swallows and spiders perform when they are building their nests and nets (Franssen, Lokhorst and Van de Poel, 2009).

Aristotle, in his work *Physics*, set the distinction that the natural things and artifacts have different properties by their nature. Technical things, that are produced with the humans' intention to service their needs, do not have the ability to reproduce themselves by inner final cause like other natural things: plants, humans and animals. This thesis has a critical value for the classification of things between the human-made artifacts and the natural things. Plato described the world as a creation by an artist and Aristotle use examples that indicate how the arts and crafts align to natural processes.

On one hand the philosophers deal with the implications of technology in the life of humans and on the other technology is an underdeveloped field that needs to be cultivated (Bunge 1979 cited in Selinger, 2012, p. 4). Technology has a markable discourse in modern philosophy. Philosophy of technology is often intertwined with the philosophy of science. The philosophical thinking about the undefined implications of the operation of an apparatus in a complex and uncertain environment is discussed in this context (Selinger, 2012). For Husserl when an object is occupied by the human senses produces the noema of the object and the intellectual procedure of thinking about the object constitute the noesis. This is a distinction that focuses on the level of consciousness. The intentional and conscious presentation of an object, that it shaped by the relation of noema and noesis in the human thought process are subjects that are excluding each other (Flores Morador, 2011). This perceptual experience and conceptual assumption of the experience in the human interaction

with the objects and the environment form the lifeworld “the world of ordinary lived experience that is in the background of scientific abstraction” (Dusek, 2006, p. 71). For Heidegger, the technology is based on the structure and function of the technological artifacts and he indicates that the objects in use are becoming transparent means by which humans interact to an environment (Flores Morador, 2011).

In *postphenomenology* technologies are multistable (Ihde, 2009). They have not a determined identity of properties but get defined by humans use in a specific context. Humans have to interpret and attribute properties to objects in order to be more than just objects that surround their lives. When humans use artifacts these become artifacts that perform a task. This purpose of use is determined not entirely by the properties of the artifacts themselves but also by the means and the ways the users deal with them (Verbeek, 2006). From this perspective the actions of humans are distinguished between the intentionality of action and its realization that transforms the *noema* to *pragma*, both objects of thinking and human’s action.

Michel Foucault, in his analysis of the technology and the power, suggested that technology has a meaning that is extended in things and practices without a predefined goal. He described this relationship as a complex of structures, design places and behaviors that govern the individuals. This domination in the life of individuals is not exercised by the law but by the normalization and not by the punishment, but by the control (Willcocks, 2004). In his perspective, Foucault made an intellectual connection between truth, the knowledge and the power. He indicated that power is not exercised from the top of the society towards the bottom. Since power passes through the structures of the society and humans are affected by the power and exercise power to others. So, the power passes through individuals and creates the reality. In this assumption of the power the involvement of technology has an important function. The extended use of technology in everyday life has a critical impact on individuals and it provides a broader term of persuasive technologies. The technological artifacts serve the intentions of the designers to guide the users attitude in a predetermined use of artifacts or in a specific way of action within the environment that they live (Verbeek, 2006).

Foucault considered also the relation between knowledge and power. He discussed these two concepts as interdependent since they are related to each other. There is a relationship of knowledge and power that indicates that any power relation bears knowledge and any knowledge produces a power relation (Willcocks, 2004). The concept of power is an issue closely attached to the phenomenon of technology. The artifacts are embedded with a moral load of intentions. Through the artifacts we exercise power over nature and also through a set of structured forms of action we also inevitably exercise power over ourselves. Following this view, technology can be "associated with diverse human behaviors, with distinctions among them, often less clear than for either artifacts or cognitions. Technological activities inevitably and without easy demarcation also shade from individual or personal into group or

institutional forms" as Mitcham suggests (1994 cited in Gerrie, 2003). In this philosophical view the technology operates as mediation between the human and reality.

The central idea in the theory of mediation is that technologies should not be understood as functional instruments, but as active mediators in relations between humans and reality. This mediating role is materialized by the specific ways in which technologies in use are present to their users' consciousness. In other words, it is the fact that technologies are not the end of human perception and action, but withdraw our attention, making vivid specific experiences and practices. Verbeek uses an example to describe this operation of technology. He suggests that when we are driving a car "our attention is not with the car itself, but with the road, the surroundings, and the traffic signs. Only when we still have to learn to drive, or when something goes wrong, our attention is drawn to the car itself. But as soon as we know how to drive and everything works properly, we embody the car" (Verbeek, 2006, p. 3). Don Ihde, as the founder of the postphenomenology, has dealt with the ideas of technological mediation and embodiment. In his works describes the functions that rely between the human intention, the body and the artifact. A person that plays music with a flute synchronizes his fingers and breaths and actually embodies the musical instrument in the process to produce the desirable sound (Ihde, 2007). Furthermore when a blind person uses a stick to recognize the things around him, although he is aware of the stick's existence, has ceased to pay attention in this artifact and guides his senses directly to the things that through the stick can be reached by his perception. With this function "embodiment or bodily intentionality extends through the artifact into the environing world in a unique technological mediation" (Ihde, 2009, p. 36). Another distinction that has a significant role in the philosophical thought about the technology is the division between the "broken" technologies and the "whole" technologies. The artifacts, as an imitation of human capabilities, can be assessed by their ability to achieve the operations that are intended to serve. When a tool or machine executes a work better than a man or than another tool is attributed as a "whole" technology artifact. From the other side, when this performance is subordinate then the artifact constitutes a "broken" technology (Flores Morador, 2009, p. 17-18). This position generates a boundary of usability of the artifact, through which an individual should face and interact with the context that is created by the terms of learnability, efficiency, memorability, errors and satisfaction. All these concepts are elements of the users' experience with the artifact and are used to the evaluation of the artifact by the users in relation to the context in use.

2.3 Conclusion

The theory of Phenomenology deals with the understanding of the world as it shows itself to us and by an extended understanding of human experience from the point of view of the person experiencing it. Phenomenology it suitable for exploring how

people design, experience and use the smart homes. The human consciousness has an intentionality that guides the human actions and the way that they experience the objects. The experience of actions or objects carries a specific meaning that depends on the humans' ongoing sense of the object (*noema*) that permits *humans' intentionality to* experience actions and objects as *taken*, something that has properties and meaning. On the other hand, Postphenomenology examines the role of technologies in human lives and it will be used to understand and interpret the impact of technologies in social, personal, and cultural life of the prospect residents in smart homes. From this perspective postphenomenology helps to interpret the way that technology engages with humans and reality and forms the *lifeworld*. Finally, philosophy of technology will be used in understanding the notion of technology, how technology *materializes morality* and the impact of persuasive technologies in humans' interaction with smart homes.

3. Smart Homes and Internet of Things

In this chapter I present a review of the literature about smart homes, human-system interaction and IoT. Finally, a conclusion is included.

3.1 Smart Homes

There are many studies of Smart Homes and the humans interactions in these computing environments. IoT has been engaged with smart homes in the last years, as suggested by many studies (Dohr et al., 2010; Michahelles, Karpischek and Schmidt, 2010). There are many types of smart homes, offered in the market, to cover the needs of users. By this vision, the smart homes are classified in three major application categories. The first category provides services to people by tracking and recognizing their actions or by detecting their health conditions. This kind of smart homes functions as information collector mechanism that supports the welfare of the residents of the home. These smart homes can be further classified into three types: a) smart homes that provide care to the seniors, b) smart homes that provide healthcare and c) smart homes that provide childcare (De Silva, Morikawa and Petra, 2012). The second category of smart homes aims at storing and retrieving of multi-media captured within the smart home, in different levels from photos to experiences. The third category is surveillance, where the data collected in the environment are used to create information that can help to actuate alarms, in the direction to protect the home and the residents from violent actions such as burglaries, theft and natural disasters like flood or fire (De Silva, Morikawa and Petra, 2012).

The users of the first category of smart homes are: a) lonely people that are unable to seek help in emergencies, b) elderly or disabled people who face cognitive or physical disabilities, c) people who need help in order to care themselves in everyday activities like eating, toileting and bathing and other critical activities such as cooking, following medicine schemas and doing laundry, d) people in villages and remote communities or in cities with inadequate health services, e) people that are involved in health care by distance or telemedicine (Chan et al., 2009). Another vision of smart homes implementation is the Energy Aware Smart Home. There are many attempts to create a system that incorporates energy efficiency features into the smart home environment. These functions are accessed and controlled through innovative solutions that serve the user-system interaction. The use of existing middleware framework like Hydra for developing the smart home environment provides the adequate infrastructure for this kind of embedded systems (Jahn et al., 2010).

In order to achieve a common ground of communication and interoperability between smart things, hardware, software of applications and humans, ontologies have been developed to support the appropriate level of interaction, integration and reuse among devices, inhabitants and external environment (Chen et al., 2009). The critical issue is how the semantic metadata is being created, collected, stored,

processed and reused in the way to serve these functions into the smart homes' environment (Chen et al., 2009).

A critical application of smart home, with important development and significant number of users is the eldercare and health care that also should be studied in this research. Many systems and projects for smart home vision of eldercare and healthcare have been developed in Europe. In the UK there has been developed a house for elderly and disabled people. It uses a sensor system that captures and evaluates vital signs and activity, and provides security monitoring and response. Also, integrates environmental control technologies for the doors, windows and curtains. In addition, there are some projects that present in an experimental level the prospect smart homes as aspects of technological optimism. The University of Ostrava in the Czech Republic has developed a smart house to investigate the human activity with infrared (IR) sensors. In France, the PROSAFE project in Toulouse has intended to support autonomous living to the seniors and disabled persons and to actuate alarms in case of emergency. This system operates with infrared sensors that are embedded in the ceiling of the house and allows the assessment of mobility and activity. The Swedish Handicap Institute has constructed a two-room SmartBo project for disabled people. The project focuses on ICTs and assistive devices and uses systems that offer to the user the ability to supervise and control functions in a home like windows, doors, locks, water outlets, electric power, cooker and bed (Chan et al., 2009).

Additionally, a separate attention has been paid to the ways the users interact and control smart homes. Some of these implementations are realized through Android OS applications for smartphones and provide users with a tracking activity technology that is composed by three layers: the smart home layer, the cloud computing layer and the application layer (Fahim et al., 2012). This application is tracking the daily activity of the elderly and reminds them any incomplete task (e.g. open refrigerator, open oven etc.), also these activities are communicated through a cloud computing to the caregivers (e.g. family members, social caregivers). This monitoring of the elderly's activities provides the caregivers the opportunity to support the vulnerable residents of the home.

Furthermore, smart homes are complex implementations that are supported, in several levels, by IoT infrastructure and constitute a part of this realm. A review of Internet of Things is needed in order to understand the interconnections and the implications of this relation.

3.2 The Internet of Things (IoT)

The Internet of Things is constructed as a system with functional structural components that work together to provide a sequence of interconnections and services. The subsystems of this structure are designed as layers that communicate successively. There is a distinction of functions between the lower and the upper layers, with an Internet layer in between to serve the purpose of a common media for

communication. The two lower layers are responsible to collect and transfer the data from the objects to the two upper levels. The upper levels are: the application layer and the middleware layer. In addition, the two lower layers are: the edge technology layer and the access gateway layer (Bandyopadhyay and Sen, 2011).

Internet of Things (IoT) is the integration between the information technologies, the internet infrastructure and other technologies, like RFID, that permit localization and tracing of physical objects. This convergence of technologies does not create a separate Internet but improves and concludes the communication of devices into an interconnected common content of objects and information. Internet of Things, as reality, completes the vision of interconnected world of things and services (Gubbi et al., 2013).

The implementation of IoT reality is a combination of double increasing efforts on the fields of network technologies and services. The vision is the achievement to provide applications with an efficient adjustment in the ICT environment that surrounds the technological infrastructure of people's life. This is a completely new approach that represents the schema of common infrastructure that can be used by applications without the need of vertical and isolated structures of networks and devices. Simply, there is a common platform that synchronizes the shared sources of the technical and physical environment (Borgia, 2014). The infrastructure is consisted of three components: a) the collection level which is the process of "understanding" the physical environment, b) the transmission level that is described as the level that uses technological structures to deliver the collected data to applications, c) the processing, managing and utilization level which deals with processing and analyzing information flows, forwarding data to applications and services, and providing feedbacks to control applications (Borgia, 2014).

Furthermore, to the above layers it is provided a taxonomy that helps defining the components required for Internet of Things from a high level perspective. There are three IoT components which enable seamless UbiComp: a) hardware made up of sensors, actuators and embedded communication hardware, b) middleware on demand storage and computing tools for data analytics, c) presentation novel easy to understand visualization and interpretation tools which can be widely accessed on different platforms and which can be designed for different applications (Gubbi et al., 2013).

The Wireless Sensors Networks (WSN) is a key component of Internet of Things infrastructure. Recent technological advances in low power integrated circuits and wireless communications have made available efficient, low cost, low power miniature devices for use in remote sensing applications. The combination of these factors has improved the viability of utilizing a sensor network consisting of a large number of intelligent sensors, enabling the collection, processing, analysis and dissemination of valuable information, gathered in a variety of environments. Active RFID is nearly the same as the lower end WSN nodes with limited processing capability and storage. The scientific challenges that must be overcome in order to

realize the enormous potential of WSNs are substantial and multidisciplinary in nature. Sensor data are shared among sensor nodes and sent to a distributed or centralized system for analytics.

3.3 Social Internet of Things (SIoT)

Beside the technological and technical infrastructure of IoT, a social perspective of the phenomenon has emerged that pays attention to IoT's influence on humans and communities. It is vital to explore and describe the boundaries that technology and especially the Internet of Things forms in relation to the society, human activities and willingness. This border can be recognized as the social side of IoT and operates as a critical factor to the overall acceptance and use of IoT by people.

Internet of Things is not an isolated technological infrastructure, but is incorporated to the society and to people's lives. There are several examples that demonstrate this statement. The aim of IoT services is to cover the needs of society. Societies are formed with the aim to provide to their members safety, shelter and healthy environment that could make them feel happier (Chen, 2012). To do this there are several paths that engage the contribution both of persons and legal entities. A sequence of procedures among the enterprises, people and government services are exploited within the IoT in the course of providing "intangible goods" such as feel special, stay healthy, avoiding unhappiness and time/money saving (Chen, 2012). The flux of the data that the objects provide and process are transformed into information through a variety of middleware software layers and protocols. This means that a person is able to use information to support its decisions in everyday life. For example the information about the traffic in the roads, the weather conditions and the technological capability to communicate with others can help a person to decide if he will attempt to move to another place by a specific route or not to move at all. On this base, the use of technology as mediation to human-human interaction or to human-system interaction has another implication related to the designers "inscriptions" of the technology objects and services that influence the behavior of users (Verbeek, 2006). In this context, IoT is examined as a persuasion technology application that embodies in this interaction a significant issue for moral reflection: the motives of the designer, the methods of persuasion employed in the technology, and the intended and unintended outcomes of the persuasion (Verbeek, 2006).

The social side of IoT has been determined by many authors as the harmonious interaction between human, societies, and smart things (Guo et. al., 2012). This vision provides the relationships that are deploying between the persons, the formed communities and the things that lead us to a more forward state of Opportunistic Internet of Things. Technology offers people the ability to sell and trade a various range of things through the opportunistic trade agents (OTA). The behavior of people, in this context, has a radical difference from the traditional trade routes. In this case the person has advanced options to act that are offered from the new

characteristics of this environment constituted by the pervasiveness, delay-tolerable, social connection and opportunistic (Guo et. al., 2012).

A framework of terms describes the social context that is formed by the use of IoT infrastructure in the context of opportunistic IoT. The terms of user awareness, ambient awareness and social awareness have a sense of fundamental components in the procedure of the "trade relationships". Both users and opportunistic IoT interact in a bi-directional way. From the one side the behavior of users affects the network and at the same time the users habits are tracked and monitored by the opportunistic IoT (Guo et. al., 2012). Also, in this case some privacy and security issues are arised. The users that allow to the device or through them to share and disseminate data are vulnerable from several lacks regarding the use of the data by the service providers, the other brokers' modifications of the messages and false handling of requests by the communication protocols.

Furthermore, the social networks are attempting an integration of IoT capabilities and services that could change the way users interact and behave to each other. There are many examples of these innovative applications that use object-generated data to provide sophisticated services such as lists of telephone calls and the ID of the callers like the application TwiPhone does in the Twitter for the members that want to provide these data to the public view (Michahelles, Karpischek and Schmidt, 2010). Also, a serial of other prospective solutions are deployed regarding the use of data that the things produce. The idea of "do-it-yourself" indicates that people can decide to collect and use data from different objects in order to create thematic/conceptual nodes of information about, for example the air pollution or city noise and to upload these data to the web. Although, this perspective of Social Internet of Things has a great value in the direction of solving problems of the citizen's everyday life and improving the conditions on the cities environment, there are some ethical issues about the privacy of the users and the question: who is setting the rules? On this point arises the dilemma about the distinction of the borders between the open persona and paranoid persona (Michahelles, Karpischek and Schmidt, 2010). It is a fact that users have adopted the idea that their privacy is a subject of conflict between the commercial use of the data and the democratic rights of individuals' and more forward to the ethics. From the other hand a defensive and refusing stance to share and communicate data, as a paranoid persona does, might be an obstacle to the creation of new services and knowledge for the common good.

3.3.1 Human-Centred IoT

As Koreshoff, Leong and Robertson (2013) suggest the HCI efforts in IoT have not been discussed with deeper human-centred perspectives of IoT like the consideration of human agency and the ability of use when multiple human interactions are occurring with multiple interconnected objects. Also, the approach of CSCW is emerging in the course of establishing the IoT as a new social reality and has a useful role in the exploratory pathways to describe and analyze the mode of cooperation

among objects, humans and space (Robertson and Wagner, 2015). There are many fundamental analogies regarding the objects seamless cooperation and the human cooperative work through the technological infrastructure that supports both the things connection and the working environments. In a more traditional frame CSCW has principles to the collaborative work that is conducted by the members of a team to achieve a common goal.

According to Schmidt and Bannon (2011):

"CSCW should be conceived of an endeavor to understand the nature and requirements of cooperative work with the objective of designing computer-based technologies for cooperative work arrangements" (Schmidt and Bannon, 2011, p. 49)

In the IoT context the objects deploy a kind of "social relationship" with the network and build an informational platform that different "working locations" engaged and collaborate in the course of achieve multiple tasks at any field that the objects act (e.g. supply chain management, traffic management, material control, etc.). Robertson and Wagner (2015) suggest about that:

"IoT applications will not design themselves, nor will they exist in 'non-places' with interconnected objects happily chatting, sensing and taking decisions. They will need people to make decisions about which sensors to use and make decisions about where these should be. Then other people need to program, install, calibrate, monitor for accurateness, clean, and repair them. And we could devote a whole paper to a discussion of the articulation work needed to embed IoT applications in everyday practice!" (Robertson and Wagner, 2015, p. 293)

Through the objects interdependencies and cooperation that consist a common practice in the IoT applications, it is identified a profound similarity with the CSCW concepts of coordination mechanisms, differences across local contexts, awareness and information spaces (Robertson and Wagner, 2015).

There is a framework of IoT perspectives that is suggested by Atzori, Iera and Morabito (2010) and describes the core ICT technologies that create the context of IoT realm and concentrates a significant number of citations by other researchers. These components are the "Things", "Internet" and "Semantics". These categories are used as topics of examination about the HCI efforts that are engaged with the IoT practices. In order to further investigation of human-centred design of HCI in IoT Koreshoff, Leong and Robertson (2013) modified and improved this framework that is consisted by an extended context of components: Internet/Things, Semantic/Internet, Things/Semantic and Internet of Things (see Figure 1). This enhanced model could be used as a thinking boundary through which IoT projects and the human-centred interaction design will be examined. This perspective of combined technologies that co-work to produce infrastructure and services for IoT

are proposed as an approach first to understand the extent and focus of IoT related efforts of HCI, second to be used as a tool to think about the considerations of HCI when design for IoT services and third to indicate the lack of engagement of human-centred perspectives of IoT (Koreshoff, Leong and Robertson, 2013). To achieve this approaching the authors reviewed the related literature and also examined the IoT related technological products that form the infrastructure of this network of objects and people (Koreshoff, Robertson and Leong, 2013). Through this investigation is discussed the way that HCI design is approached in IoT devices and services.

The focus on human agency and how people anticipate the technology is not based in a deep human-centred ground and also in the perspective of human flexibility and diversity when multiple connected objects interact with many users.

Koreshoff, Robertson and Leong (2013) suggest the use of PD in order to design a human-centred IoT:

"One approach to ensure a human-centred focus in the evolving agenda of the IoT is to draw upon Participatory Design. This is because Participatory Design has an inherent concern for co-designing with people, encouraging people to imagine and shape how technology can best work within their lives." (Koreshoff, Robertson and Leong, 2013, p. 341)

Participatory Design provides the methods and tools to reveal the needs and desires of people, and makes the design of devices and services of IoT more effective and attractive.

Koreshoff, Robertson and Leong (2013) state about their proposed PD contribution in the field of IoT human-centred design:

"With rich traditions of ethnographic research and deep understandings of co-design, through workshops, iterative prototyping, and many other design methods and tools, Participatory Design is well suited to support the development of future IoT technologies that are human-centred and respectful of their needs and those of their communities." (Koreshoff, Leong and Robertson, 2013, p. 342)

All these guide us to understand the framework as a design perspective and to think about the HCI issues when we approach the design human-centred IoT devices (Koreshoff, Leong and Robertson, 2013).

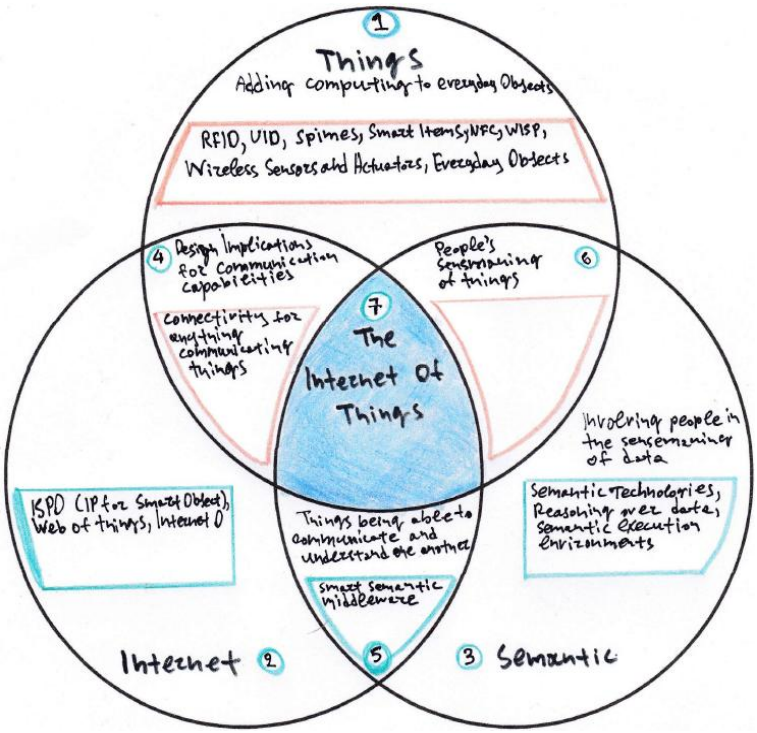

Figure 1: Modified version of Atzori et al.'s (2010) 'Internet of Things' paradigm (Adapted from Koreshoff, Leong and Robertson, 2013)

In this direction a method of current projects research is proposed, as presented in literature, regarding three components a) evaluating of design ideas, b) exploring systems through design and c) exploring technical components through design (Koreshoff, Robertson and Leong, 2013). This review is extended in commercial products that are used both to support people and home functions. These artifacts are divided in person-centric and home-centric products and form different context of interaction. The person-centric products like sleep, fitness or weight monitoring devices are based in wearing the device or in the case of a scale device, in standing on its physical interaction. Another way of interaction with these objects is through a smartphone application. In addition, the home-centric artifacts such as the door sensors and actuators and alarm systems use mechanisms to gather data from the environment and interchange them with other devices. So, these devices are more interconnected with each other in order to establish a network of functions within the smart home, but this does not exclude the interaction with humans as this is utilized through the display controls or move tracking sensors (Koreshoff, Robertson and Leong, 2013).

Some projects suggest the interaction of humans with smart home environment through the creation of a "virtual housekeeper" that imitates the characteristics of

residents by perceiving the personality model of the user. This emotional virtual human management system is designed in order to provide a friendly communication with the user and to achieve a comfort living environment (Du and Wang, 2012). The creation of the virtual human follows these steps: first the user has to select the virtual personality, second the system recognizes some critical characteristics of the user such as the voice and face and then the system starts to detect the emotional status of the user, third the perceptive level of the system processes the reaction of the user and compare it with the knowledge base in order to stimulate the mood of the user. Finally the system achieves to imitate the emotional state of the user by the perception of emotional, mood and personal characteristics that have been recognized by the sensors. With this information the "virtual housekeeper" configures the appliances and functions of the home according to the user's preferences in the current time.

Another perspective for the human-system interaction is deployed by a relational ontology where smart homes and residents are not pre-given or interdependent factors but they come into existence in intra-actions in material-discursive practices (Finken and Mortberg, 2011). The humans and non-humans are intertwined in these smart homes and moreover the services and activities are in a continuous process of evolution according to the people's requirements and the technology's solutions.

3.4 Conclusion

This chapter gave a description of the literature about the Smart Homes and the Internet of Things as two interconnected realities that collaborate to produce services to people. Also, other perspectives of these technologies were reviewed such as the Social IoT and human-centred IoT that are useful concepts in order to interpret the human-system interaction phenomenon in smart homes.

The review of literature showed that the human-computer interaction research about IoT pointed out a lack of a strong human-centred perspective. The use of Participatory Design is proposed to provide a significant contribution in order to establish a human-centred design of IoT (Koreshoff, Leong and Robertson, 2013). Also, the importance of human agent decisions in the process of design, programming, monitoring and repair of IoT applications is projected in the literature (Robertson and Wagner, 2015). I intend to use the human-centred interaction approach in IoT and smart homes in the course of the discussion of empirical findings. Also, the literature of IoT and Social IoT will offer important probes on the process of research conduct, data analysis and discussion of empirical findings.

4. Methodology

In this chapter I describe the research paradigm to locate my research. Then the data collection methods and data analysis are presented. Finally, validity, reliability and ethical issues are presented and discussed.

4.1 The Research Paradigm

I decided to use an interpretive research paradigm that is based to the philosophy of postphenomenology that I used in the theoretical framework, because postphenomenology has its origin, as Ihde (2009) suggests, in the phenomenology and phenomenology is located in the interpretive worldview (Burrell and Morgan, 2005). Although, an interpretive research can be critical (Walsham, 2006), there are some fundamental components that distinguish interpretive and critical research: choosing a style of involvement, gaining and maintaining access, collecting field data and working in different countries. The way that the researcher conducts the above stages of the study, characterizes the research approach that he follows.

I use the *phenomenologically-situated* paradigm of HCI in order to conduct this research (Harrison, Tatar and Sengers, 2011). I follow this as an appropriate research paradigm to investigate the human-system interaction in smart homes. This paradigm provides a set of perspectives that helps the understanding of humans interaction with the computerized systems. These perspectives are: dynamic use context, socially situated, indirect and multiple goals, non-task oriented computing and handling emotions (Harrison, Tatar and Sengers, 2007).

As an interpretive researcher, I ought to proceed in an interpretation of what participants of the research experience without the intervention of my own background, context, and prior world's perceptions (Creswell, 2009). Also, the qualitative research approach helps the researcher of Information Systems phenomena to understand the experience of the participants in their context and gain a deep insight into these implications (Klein and Myers, 1999).

4.1.1 Research Approach

This research follows the *qualitative research* approach that is posed in the interpretivist philosophical worldview, especially the phenomenological and postphenomenological, and is intended to investigate and understand the experience of the participants, in the social and organizational context of smart homes. The decision to use a qualitative research approach was taken because I wanted to shed light and understand people's experiences in smart homes. Designers and workshop participants have contributed with their experiences in this research. The qualitative research helped me to interpret the meanings that they were attached to these systems, such as the way that they perceive the smart home functions, the way they interact with the devices and the space and the meaning that they give to the artifacts. It is a critical distinction that makes the qualitative research appropriate for this study and

this is the ability to offer an in-depth view of the particiapnts thoughts and actions, in contrary to the quantitative study that is useful for falsifying theories (Flyvbjerg, 2006).

4.2 Data Collection Methods

The research was conducted between April and June 2015. The data of the research were collected with three different methods: interviews, workshops and observations. First, the interviews with the designers and implementers were conducted. Then, two workshops with prospective residents were held. And, I had also the opportunity to observe the Smart Home systems in use from the interviewees that demonstrated me the functions of their technological infrastructure.

4.2.1 Research Setting and the Participants

This research was conducted in the region of Attica (Greece). I contacted one (1) company that operates in the market of smart homes and IoT. I had four (4) interviews with staff members of the company that work on design and implementation of smart homes. The time period that the interviews were conducted was April-June 2015. Further, I conducted two workshops with eight (8) participants. The workshops were conducted in the same period of interviews (April-June 2015) in Athens.

The number of designers was four (4). They work in one company where they contribute to the design and implementation of smart home environments. They were interviewed in their offices. The following table presents the participants involved in the interviews:

Table 1: Interviewees

Name of Participant	Status	Age	Participation status
Manolis	Electric engineer	48	Interviewee
Spyros	Automation engineer	29	Interviewee
Kyriakos	Sales director	50	Interviewee
Ilias	Customer Key Manager	33	Interviewee

Here, I should clarify that the concept *designer* is used to cover all the occupational groups that are involved in the design and the implementation of smart homes. Further to the designers, eight (8) ordinary persons also participated in the workshops. Three (3) persons participated in the first workshop and five (5) persons in the second workshop. They were invited because in my attempt to get access to the clients of the company that were already living in smart homes it was impossible to get the contact information from the company. In order to overcome this research gap I decided to use the method of workshop so I invited persons to participate in the workshops. The workshop's participants pretended to represent the prospective residents. The participants are presented in Table 2.

Table 2: Workshops participants

Participant's name	Status	Age	Participation status
Petros	Working/Private clerk	28	1st Workshop participant
Nikos	Working/Private clerk	25	1st Workshop participant
Maria	Working/Private clerk	29	1st Workshop participant
Stavros	Working/Public servant	37	2nd Workshop participant
Dimitris	Working/Public servant	55	2nd Workshop participant
Thanos	Working/Public servant	58	2nd Workshop participant
Thalia	Working/Public servant	54	2nd Workshop participant
Lena	Working/Public servant	42	2nd Workshop participant
Paraskevas	Working/Public servant	37	Coordinator of workshops

This table describes the participants, their actual names are changed in order to protect their privacy and to avoid harming them. The workshops participants' occupational background and age are also included in Table 2.

4.2.2 Interviews

The research was conducted between March and June 2015. The interviews with the designers of smart homes offer the opportunity to study the research problem in its context. I decided to use semi-structured interviews due to the fact that I wanted to

examine the experience and the views of the participants about the design of the infrastructure, the prospect residents' needs and the ethical issues that emerge in the framework of smart homes implementation. In the course of contact with the interviewees, I asked them for a possible appointment and I ensured them that I would be able to visit the company's office at the time it would be convenient to them. As a researcher, I prepared the interviews considering the research questions and I contacted the participants in order to investigate their willingness to participate. In this attempt I faced some difficulties to gain access to the companies and from ten (10) requests to companies only one (1) response was positive. The participants were all members of the same enterprise that offers smart homes infrastructures.

The interviews started with a presentation of me and the purpose of my study. Furthermore, I introduced the participant to the research context and I asked about the company, the field of activity, in addition I asked the interviewee to read and sign the Informed consent form as an acceptance of participation in the study, see Appendix A. The interviews were recorded after permission in order to avoid any misunderstanding, missing words or lack of important sections of the interview. The argument to record is that note-taking is time consuming and distracting (Crang and Cook, 2007, p. 42).

After the interviews, an email was sent to the interviewees to thank them for their participation, but I didn't send them the transcribed interviews to the participants because I had informed them that they had the right to withdraw from the study, to have access to the recorded and transcribed material and to ask for the removal of their data any time they wanted without giving any explanation. No one of them asked access to its own data. The interviews were transcribed verbatim. This transcription was made for the four interviews through the transfer of files from the digital voice recorder device to the PC. The procedure of transcriptions was a time consuming task, due to the fact that the participants expressed many ideas and feelings.

4.2.3 Workshops

Workshops were another data collection method used in this study. Individuals were invited to represent the smart homes users. Prospective residents are the subjects of Human-System Interaction within these technological environments. Due to the company's denial to provide me access to their clients, I decided to conduct workshops with individuals from my network-work and studies-environment in order to fill the gap.

Workshops are an essential method of Participatory Design, a design approach that was born at the Scandinavian countries. Participatory Design interconnected the fields of art, technology and politics in order to democratize the design and establish the artifacts and structural interventions in the citizens' life (Bannon and Ehn, 2012).

The first workshop was held with three participants. The workshop was conducted on 25th of April 2015. The participants were friends-classmates in the master

program and their ages were between 25 to 29 years old. The workshop took place in a café. This place was used to help the participants to feel comfortable and to imagine themselves in the smart home that was demonstrated in the videos. In addition, this was an environment that could help all of us to feel more comfortable with the aim of the workshop.

The second workshop was conducted with five participants on 8th of May 2015. All the participants were colleagues from my work and represent persons at an age range between 37 to 58 years old. The workshop took place in the office after working hours. This was the most efficient way in order to achieve a common convenience time for all. Both workshops were recorded with an audio recorder device. The recorded data were transcribed verbatim. I participated in the workshops as coordinator.

The structure of the Workshops was based on three short videos that demonstrated the functions in a smart home and the possible uses by its residents. These videos derived from a research in YouTube and are:

1) Living Tomorrow house of the future.
https://www.youtube.com/watch?v=9DJr8QwgLEA

2) My home automation system in an apartment.
https://www.youtube.com/watch?v=tOZHMNfG2kM

3) What is a Smart Home? Video Example of Smart Home Technology in Action.
https://www.youtube.com/watch?v=i73n-LTXPIM

The presentation of these three videos was followed by a conversation about the use and feelings of the participants if they had the ability to live in this environment. The conversation was built upon a *scenario* that has a linear form of “happenings” of a usual day’s activities in a family (see Appendix C). Workshops were conducted as an alternative way to reach residents of the smart homes, because the company that I collaborated with was negative to provide me with the contact information of their customers. The workshop participants were invited to act as an added source of research data that simulated the “real use” of the smart homes by prospective residents with an advance of “imagination” and “desires” that were expressed from the participants.

4.2.4 Observations

Observation was a third data collection method. I had the opportunity to *observe* some of the smart home facilities offered by the company where the interviewed designers work (Walsham, 2006; Wagner, Bratteteig and Stuedahl, 2010). Examples of services were: lighting control, adjustment of heating, configuration of shading, etc. The company that I visited to conduct the interviews has installed in its offices

automation equipment, like sensors and actuators, dimmers, thermostats, displays, etc. I asked the interviewees to show me the functions of the system. In my observation notes were kept in order to use them in my data analysis. I respected the will of the interviewees not to take any photos.

First visit/observation: 15 April 2015. I observed functions that were demonstrated to me by the interviewee like the "exit scenario" with the activation of an alarm system and the close of doors and windows and other functions like the temperature adjustment and the interface control of the system. The time of the observation was approximately 20 minutes.

Second visit/observation: 22 April 2015. I saw the functions of the smart home such as the monitoring of the room, the climate condition configuration, the entertainment system and the control of lights and heating. The time of the observation was approximately 20 minutes.

Third visit/observation: 19 May 2015. I observed the control of the house from a smartphone interface that depicted the plan of the rooms and the spaces. All the functions was represented in this plan. The time of the observation was approximately 15 minutes.

Fourth visit/observation: 1 June 2015. This observation included a smart home in a permanent exhibition at the company headquarters. It simulated a home with both external and interior spaces. The functions, like the opening and closing of the garage door, the lighting control and the configuration of the internal and external shading were all activated by a mobile application without the use of several controllers and any other special devices. The time of the observation was approximately 30 minutes.

The opportunity to see the functions of smart homes, in an installed technological environment with an infrastructure of sensors, actuators, cameras, dimmers, etc. was useful to me to understand the designers' point of view and the prospective residents' perception of technology (Blomberg et al., 1993).

4.3 Data Analysis

The recorded interviews and workshops were transcribed with the use of a computer software. Also, spreadsheets were used to code the raw data of the empirical material.

About phenomenological analysis Lichtman (2013) states:

"If you were following a phenomenological tradition, you would be interested in the lived experiences of the individuals. You will need to explore some of the philosophical underpinnings of phenomenology, and your data analysis would be

facilitated if you bracketed your views. You might choose to conduct either narrative analysis or thematic analysis." (Lichtman, 2013, p. 257)

The data analysis procedure has to go back and forth across the data in order to reach their meaning (Lichtman, 2013). In this direction, I carefully listened several times the interviews and workshops. I went back and forward across the texts and audio recorded material to achieve and interpret the original word spelling of the participants and the meaning of their speech. I preferred to follow the thematic analysis as it is described by Lichtman (2013, pp. 252-255). The procedure is as follows:

Step 1: Initial coding. It goes from responses to summary ideas of the responses

Step 2: Revisiting initial coding

Step 3: Developing an initial list of categories

Step 4: Modifying initial list based on additional rereading

Step 5: Revisiting your categories and subcategories

Step 6: Moving from categories to concepts

In this procedure I read the raw data of interviews, workshops and observation in order to define Codes, Categories and Concepts. My initial coding produced 188 codes for interviews, 184 codes for workshops and 36 codes for observation. After removing duplicate (differently named, but with the same meaning) and redundant codes, the number was reduced to 76 codes for the interviews, 163 for the workshops and to 14 for the observations. These codes from the three raw data sources were merged and sorted into 69 categories. Finally, the categories were condensed into 5 themes: Design of Smart Homes, Home Control, Interaction with the System, Ethical Issues and Awareness and Considerations.

4.4 Validity and Reliability

The validity of the research is ensured through the accuracy of the findings by using validated test procedures and the reliability of research is certified through the comparison with different researches and projects (Creswell, 2009). In addition, attention was paid in the cross-checking of codes and the transcription process to avoid any mistakes. The obvious solution to ensure the research's validity and reliability is to keep consistent to the codes of the engaged parts (researcher, participants) and other researchers' work in order to communicate a common language of codes and meanings.

In order to ensure the validity of the research I followed some strategies as indicated by Creswell (2009):

a) The triangulation of data: I used data that were collected from different sources like interviews, workshops, observation in order to identify themes

b) Present negative or discrepant information that runs counter to the themes: I pointed out any discrepant information.

c) Spend prolonged time in the field: I spend a prolonged time in the research field, due to the fact that I conducted four interviews, two workshops with twelve (12) participants, and I observed smart homes services and functions and the related equipment.

In order to ensure the reliability of the data that were obtained, the interviews and workshops were recorded and transcribed verbatim and then read through once in order to check for potential mistakes made during transcription. The used codes were constantly checked and compared, to ensure that there was no drift in their definition during the process of coding (Creswell, 2009).

4.5 Ethical considerations

The research arises some ethical issues about the collection of data, the contact with the participants and how to handle the examined documents. First, the privacy of the participants was protected and the sense of confidence between the researcher and the participants was built. Also, issues such as identification of the purpose of research, identification of the benefits for participating, assurance that the participant can withdraw at any time and the identification of the researcher was resolved by an agreement between the participant and the researcher in the form of informed consent (Creswell, 2009). These issues were extended in this research because there was the opportunity for observation of the system in use in the offices of the interviewees and I assured them that this research will not harm them.

5. Empirical Findings

In this chapter empirical findings will be presented. The experiences of the participants with the smart homes are provided and the themes, that my analysis yielded, are described.

The empirical findings will be presented in this chapter. The empirical material was analysed with a thematic analysis and the extracted themes were:

1) Design of Smart Homes
2) Home Control
3) Interaction with the System
4) Ethical Issues
5) Awareness and Considerations

This table describes the themes that were extracted and the subcategories that constitute each theme.

Table 3: The five themes and the subcategories

Main Category	Subcategories
Design of Smart Homes	*Requirements specification, Clients needs, Perspective of the Internet of Things, Clients information*
Home Control	*Easy Living, Cross-platform interface, Simple interface, Home automation, Notifications by the system, Free of commitment of the physical presence, Distance control, Safety and security, Customization*
Interaction with the System	*Technological mediation, Loss of remembering, Change of habits, Support of living*
Ethical Issues	*No collected data, Privacy protection, Companies protect the system, Sense of loss the personal contact, There is no trouble to do things, System intervention to the humans' relations, Fear of breach, Legal concerns, Refusing to be replaced by the system, Data are collected and stored-big data*

Main Category	*Residents' awareness of the services, Improve the life, Limitations in common space, Human factor is the cornerstone of systems, System dissemination involves vulnerability, Use of out-of-scenario knowledge*

Each theme including the subcategories will be presented in the following subsections. They are presented in the order they are described in the table.

5.1 Design of Smart Homes

The theme Design of Smart Homes is constituted by the subcategories:

Design of Smart Homes	*Requirements specification, Clients needs, Perspective of the Internet of Things, Clients information*

The implementation of a smart home takes into account several factors such as the infrastructure, the equipment and the needs of the residents and the environmental and technological factors.

This is described by Manolis, in the following way:

"As a company we use certified stuff and procedures… I take account of the requirements specification on the installed system both at the level of functions and the use of the equipment."

and he mentioned also:

"I' am responsible to study the needs of the clients and to offer the clients comfort, energy saving and security"

The designers study about the *requirement specification* of the equipment. The interviewed designers explained what they consider about the residents and their needs. Hence, the designers communicate with the prospect residents to decide the type of the infrastructure that would be useful to install. They use inteviews or surveys to identify prospective residents desires to propose a set of equipment and functions that applies to the residents wishes. Furthermore, the designers take into account the equipment, the interoperability of the devices and the environmental conditions.

This is clarified by Ilias:

"We have the part of motors and we have the part of automation, receivers, sensors and controls. After we can divide the technologies on the base of the market, we talk

for garage door, for roll or for internal shading, lighting, industrial roll, commercial building solution, facade. We have two categories: the wired and wireless and then the advanced wireless as we describe as RTS and beyond this we have the hardware that is separated in automation and mechanisms."

The design of smart homes depends on the aims of automations that will be installed. The devices that control the garage door, the lights and shading or functions as the energy management and security are factors that modify the infrastrucutre of the project and the implementation of smart home. Although the type of infrastructure, wired or wireless, and the devices that the designers use are critical for the implementation of a smart home, there is another factor that they also should pay attention, namely the *clients information* about the services and functions of a smart home. Further, it is also important for the designers to know the *clients' needs.*

This is described by Spyros:

"We have discussions with the company that constructs the building and we are discussing about their needs and also we present to them the options and the solutions that we can offer. The building is a complex of three apartments, so we focus on the design on prospect residents and we align the approach of installation to them."

The designers inform their clients about the functions and the services of smart homes. On the other hand the workshop participants express their views and desires about the services and functions of smart homes. There is a common approach at the side of the designers. That is to offer high standard equipment to the direction to build a smart home environment that will facilitate any particular need of the residents.

In addition, workshop participants expressed the view that the technological infrastructure could offer some extraordinary services like cooking or watering of plants. The workshop participants imagined the possibility that a "smart" oven could cook the food without their supervision. Also, the automated watering of plants was another idea that they expressed. These thoughts, while not too far for the technological capabilities, describe an assumption that a smart home can replace the human everyday tasks. The designers present to the prospect residents the equipment and the functions that they offer in order to achieve a level of communication with them.

This is described by Kyriakos:

"The customer often does not know which are the capabiliities of the system, we explain to him and he says "Oh, yes. We will do it."

Further, they discuss with the prospect residents the aim and the scope of the installation. Sometimes, the designers return to the prospect residents and propose

changes to the installed services when the needs of the prospect residents or the budged are modified. The process of the design and installation of smart homes is always ongoing. Finally, there is a convergence among the designers vision of the smart home design and the prospect residents needs and desires that create the infrastructure and the services of installation.

The *perspective of the Internet of Things* is also what constitutes the theme design of smart homes. The installed infrastructure of smart homes like sensors, actuators, cameras, thermostats or dimmers are interconnected and communicate through the central server of the home. This communication is extended to the IoT through the web and the middleware level of the IoT that permits the interoperability of software and smart objects (Azzara et al., 2013).

This interconnectivity is clarified by Ilias:

"We connect the devices and then the user may control these by smartphone or tablet. Any device has the ability to connect with the web, through the central server, so the automation that we realize is part of IoT or could be part of this web. Yes, we are considering these projects as a common infrastructure that support web applications and cloud computing."

Thus, the residents who control the smart home devices through the embedded displays or smartphones, actually use the IoT infrastructure.

5.2 Home Control

The control of the house is another theme that emerged out of the analysis of the empirical data. The theme is constituted by subcategories:

Home Control	*Easy Living, Cross-platform interface, Simple interface, Home automation, Notifications by the system, Free of commitment of the physical presence, Distance control, Safety and security, Customization*

Easy living covers a wide range of elements included in a smart home. This is expressed in the designers' visions of what should be included in a smart home. The workshop participants emphasized also to the importance of easy living. For example, they focused on the system ability to close and secure the windows and doors without the efforts of the residents. Also, the control of heating and the energy saving management of the home were annotated to workshop participants' statements. They wished life in the smart homes to become more comfortable compared to this of a traditional home. They would like to have a warm environment that is self-configured in order to provide them the ideal conditions. This is expressed by Nikos, one of the workshop's participants:

"It seems difficult in the morning. But when I know that all the conditions are perfect to take a bath, the temperature, the water in the bathroom is warm, I would do it much more easily, but now I do not really do it. But if I had these possibilities I would do it. Yes, and I would like to drink two or three sips of coffee at home, to eat something, but I really don't do it because I don't have time. I'm bored, I'm bored."

Regarding the *home automations* and *notifications of the system*, the designers claimed they offer a home control through web, smartphones, tablets or PCs. In the smart home the residents are able to use the embedded displays and other devices such as dimmers, sensors and thermostats to control the home functions. The designers emphasise the *cross-platform interface* as another control element included in a smart home. The user's interface in the smart homes is the same at any device that the residents use to control the smart home with. This is explained by Spyros:

"The same interface is available from smartphone or pc in every place through the web. The user has to touch on the screen control to configure whatever she/he wants."

The cross-platform interface may also facilitate the training of the user in its use and further provide an easy mean of controlling the home in any place and at any given time. Furthermore, there is an additional component to control the home with namely customization of settings and functions. The designers offer the residents, through the technological infrastructure, the capability to set preferences and settings in the home functions.

This is described by Kyriakos:

"Because, we can have a Greek menu and a domestic worker who works at home or care the kids may do not understand the Greek control interface and wants to turn it in English. We are interested in that the control interface we use can be altered into Greek at any time and moment."

Hence, the customization of the functions permits the residents to deactivate the alarm when they are at home or to set the lights that stay lit at the night. This is a level of customization that helps the residents to control the home functions in order to save energy and to achieve a sense of security and convenience. This may also facilitate the lives of the elderly people that need special facilities at home (De Silva, Morikawa and Petra, 2012).

This is obvious in the words of Petros, one of the workshop participants:

"For example, in case someone who has mobility problems and has to go in the rooms to close the windows or the lights, it helps there too. We also know that if a resident does not have the mobility skills, we can say an elderly, needs some things to be done automatically."

In addition, there are two other interdependent categories in the theme home control: *free of commitment of the physical presence* and *distance control*. The provided ability to control the home functions from anywhere, through the infrastructure of Internet, extends the boundary of interaction between residents and home. This is described in the words of Ilias:

"So, this function is giving the comfort and the very important capability of safety that there is in combination with the products, hardware and software of installation, that it is giving the capability of safety and security, by distance always, and the prevention of invasion or the added anti-burglary safety and also energy saving. From the smartphone, as a multichannel controller, the user can handle every device… I can activate a device like a coffee machine or a boiler or configure the thermostat in the lower level of the home, or activate or deactivate a zone of alarm in my place."

Furthemore, *safety and security* are other issues included in the theme Home Control. The importance of safety and security is described by Stavros, a workshop participant:

"Always we start from the priorities. It is good to make your coffee, but it's not necessary. The security issue is for me the most important, from the check of the woman that babysitting your child until a burglar. However, all these things are not much needed and some are immaterial. The issue of security for example, I think that is the maximum. Either we talk for the children safety or the privacy and property's protection. Now if you open the blinds and windows it is good but it's not the most important."

5.3 Interaction with the System

The theme is constituted by the subcategories below:

Interaction with the System	*Technological mediation, Loss of remembering, Change of habits, Support of living*

Residents in a smart home use functions and services that may facilitate their everyday life. The presence of artificial objects such as sensors, cameras, dimmers, displays and many other things constitute a smart home in IoT. The functions of smart homes are realized through the technological infrastructure that is installed in a smart home.

This is described in the words of Ilias:

"I can receive information about a bulglary or damage in the house and I can check the pumps with my smartphone, if I have a home in a village and I don't have time to visit it or I visit it only in the summer."

Technology, in this case, forms the visual depiction of the constructed environment that constitutes the house. Furthermore, it represents the occurrences at home in a set of information. The user is able to intervene from a distance at home that is represented on the display. The device that controls the home withdraws the attention of the user. The user pays attention, through the mediation of device, to the home and to the decisions that are appropriate to take.

The use of default functions that are set by the designers and the prospect residents contributes to the phenomenon of technological mediation in smart homes. The automated configuration of heating by an embedded thermostat or the preparation of coffee by a coffee machine are some services that the workshop participants would like to use.

This is described by Thalia, she said:

"I wake up at 7 that means that the heating must have already begun working. I would like, however, when I wake up, the home to be at a good temperature condition. Also, I would love to prevail a soft light, the light should progressively strengthen, so my eyes to adjust to the light. It could open the TV because I may want a very quick briefing of news and weather which is also necessary to have and a coffee to be prepared."

Furthermore, through my observations regarding the use of smart homes, I understand that the residents may interact with the devices, from an interface on a display that simulates the home settings and spaces.

Another interaction to the system is described by Spyros, one of the interviewed designers. He claimed that smart homes improve the life of residents through a simple interface in smartphones. This is an embedded display in the house via web, wherever the resident is. The management of energy, the lighting control and the surveillance system are key components that are offered in these homes.

This is described by Spyros:

"The services such as the energy management, the surveillance system or the lighting control are some solutions that can save time and efforts for the residents. I believe that the home that we create is a place that improves the life of the residents because it is designed to help the everyday life of them."

The interaction is utilized through a complex net of interconnected sensors and actuators that gather data about moving, air condition and absence or presence of residents in a room. Thus, the designers of smart homes utilize a system that offers

users the ability to intervene in the home devices operation. The home system is becoming a part of the workplace or any other environment that the residents are in, because most of the residents bring with them smartphones or work with a PC that offers connectivity to the web, so there is the ability to monitor their home with these devices. This is a fact, because the resident has a permanent connection with the condition of the home through the smartphone, tablet or PC and this creates an added factor that she/he should pay attention to when is on the work.

The *loss of remembering* is another element included in the theme Interaction with the System. The workshop participants imagine that smart homes offer them the chance to delegate tasks to the system. In a smart home -as a computing environment- the resident does not need to consider remembering whether the oven or the window is closed, the smart home does it for her/him.

Maria, a participant in the first workshop, said:

"Many times I'm going from room to room to close the windows and I'm returning back to check the windows one by one because I forget if I have closed them. The most important of all... is to automatically close windows and doors when I leave the house and to be sure that they are closed."

In addition, the *change of habits* is a critical component in the interaction with the system. Change of habits was expressed both by the designers and the workshop's participants. The offered services in a smart home create an environment that either offers the residents the opportunity to change their habits or forces them to change their habits. Kyriakos, one of the interviewed designers, referred to the design and implementation of smart homes that, as he believes, provides conveniences and functions that facilitate the residents to change their habits. He said:

"These are functions that exist and are requested by customers. The parents, for example, can control the home when the children return from school in the afternoon. They can activate the external alarm, disable the internal motion sensors and if the bell rings they can see from their smartphone if they will open the door or not."

The control and monitoring of a home from distance is a service that gives people the capability to provide access to their homes at any time and from anywhere. Due to this function when a resident is at work and the kids are coming back from school they don't need to go home to open the door. Since this can be performed with a smartphone or tablet and always by distance. Although, the kids can have a key to open the door, but the parents may feel safer when they control the access at home. The system interacts with the residents and sends alarms to them about the presence of a person in the entrance, through the cameras and the moving tracking sensors. So, the way that the residents control place and time entirely changes compared to

traditional home. Also, the workshops participants, as prospect users, imagine some cases that the use of the system may change their habits.

Nikos described this:

"I leave from home and I never activate the alarm, because I'm bored. Because to close shutters, to put alarm, takes me time. If this task is performed by a system I would put them every day. It is very good."

The workshop's participants express that residents' use of services in smart homes may change their everyday life's habits. For example, parents need not to be present at home when the children are coming back from school. They can open and secure the home by distance.

The *support of living* is an additional aspect that constituted the theme Interaction with System. This was presented both as an aim of the designers and as a desirable service by the workshop's participants. The designers asserted that the life of the residents might be supported in a much greater extent than they are able to imagine.

This is described by Kyriakos:

"The life of the residents is supported in a greater extent than they imagine, saving them money and time. Because, when you leave from your home you have to close eight windows and put the alarm and on the other hand is better to have to activate only the alarm and automatically to turn off the lights and to close the windows and you can leave within 30 seconds. Also in the evening if you are going to fall asleep on the couch, the system is turned on sleep mode and the windows close, the door is locked, the lights of the day turn off and left some lights on and the alarm is activated."

The ability that the smart homes offer to save time when you close and secure the home and to save energy when the lights or the heating are self-configured by the system are evidence of a supporting automation environment. The devices that smart homes use implement these services through a process of orders that take from the central server and actuate motors to perform the actions.

5.4 Ethical Issues

An overall view of the subcategories that constituted the theme ethical issues is provided here:

Ethical Issues	*No collected data, Privacy protection, Companies protect the system, Sense of loss the personal contact, There is no trouble to do things, System intervention to the humans' relations, Fear of breach, Legal concerns, Refusing to be replaced by the system, Data are collected and stored-big data*

This theme is formed by a net of issues related to the fears that the workshop's participants expressed concerning privacy and the possible implications of the system in their life. The designers assured that they protect the residents' personal life and provide a security level to the smart home installation that not permits the breach of the system.

There is a relationship between the term of *no collected data* about the use of smart home by the residents and the belief of the workshop participants that *data are collected and stored-big data.* The designers claimed that they don't collect data about the habits or the personal life of the residents, and on the other hand the workshop participants expressed concern about the technological ability of every company to collect data. This is described by Lena, a workshop participant:

"The companies keep details for everything, even if they are bound by the law to protect personal data or privacy."

On the other hand Spyros, an interviewee, said:

"We don't keep data for any home or the residents personal habits. We stay away from the residents' life."

Furthermore, the *protection of privacy, companies protect the system* and the *fear of breach* and *legal concerns* are another set of subcategories that constitute an expression of different views between the interviews and workshops participants. Both sides claimed their own concerns about the protection of privacy and the fear of an unauthorised access to the system.

These issues are described by Spyros, one of the interviewed designers:

"We ensure the reliability of the system and the protection from abuses by third party and that no one can intervene in the lives of residents. The problems that are arising are about the safety of the software."

This argumentation continues with the workshop participants to concern about the *sense of loss the personal contact* and *the system intervention to the humans' relations.* They want to do things like the treat of their friends and refuse to delegate the smart home system to do that. This is described by Stavros, a second workshop participant:

"I will not expect the system to prepare the coffee or to watch the child from the camera instead of going to his room. No, I would not want the system to intervene in my relations with my friends. I want to treat my friends."

Finally, the sense of the workshop participants that *there is no trouble to do things* and the *refusing to be replaced by the system* are subcategories that produce a moral perspective about the sense of humans as entities that are threatening by the technological environment that they live. This described by Thalia:

"We are sitting and talking and you are looking us, for God's sake I do not know if you already have recorded it, about unecessary and ridiculous things. For us, the life is more much complex to do any of this."

Their emotions and bodily skills are subjects of a "reappraisal" from the view of smart home system and the designers *embedded morality* (Verbeek, 2006) in the artifacts and functions. There is a view from the workshop participants that reveals their angle for the smart home services that accused as "replacing" of their bodily skills and activity and furthermore acknowledge the "gap" between the older people and their attitude with this technology.

5.5 Awareness and Considerations

The identified subcategories of the theme are:

Awareness and Considerations	*Residents' awareness of the services, Improve the life, Limitations in common space, Human factor is the cornerstone of systems, System dissemination involves vulnerability, Use of out-of-scenario knowledge*

Awareness and Considerations were identified in the workshops. This theme is constituted by the expressed perspectives and visions of the workshop participants about life in smart homes. Workshop participants assumptions are expressed with a deep discussion about the limits and the vulnerability of the system and the role of the human in this home. The workshop participants seem to be aware of the abilities of home control, home monitoring and control of appliances that are offered by the system. This forms a use context for the workshop participants that is based on a

positive stance about smart homes and how the included technologies may improve their life. The participants seem to be aware of the functions of a smart home.

Awareness and considerations were illustrated by the participants focus on aspects such as *limitations in common space, human factor is the cornerstone of systems security, system dissemination involves vulnerability* and the *use of out-of-scenario knowledge.* The workshop participants are aware about the limitations of the smart homes' scenarios when the people share a common place. For example the turning on the lights in a bedroom may distrube the person who sleep when you are already wake up.

This *limitations in common space* is expressed by Stavros, regarding to a "wake up scenario", that is included in the workshop scenario:

"*Sorry, if you are in a common bedroom with your wife, this cannot be done"*

This awareness seems to be useful when the workshop participants talk about life in smart homes. Also, the workshop participants believe that *human factor is the cornerstone of systems security*. The key factor for the security of the smart home is the human agent and her/his actions or omissions. They have previous experiences that bring into this discussion. This is a *use of out-of-scenario knowledge* such as the argument that a member of the company that maintains the alarm system of the home is the first suspicious person in a case of a burglary.

In addition, another issue pointed out was that the *system dissemination involves vulnerability.*

This aspect is described by Petros:

"But, if it is so widespread this technology, doesn't this increase the risks? Could anyone interfere and control the home?"

The workshop participants expressed worries due to the technologies included in smart home. The system dissemination increase the danger of breach or "hacking"and the exposition of the personal data of residents (Guo et. al., 2012).

Although, the above worries have a negative impact on workshop participants overall view of the smart homes, they expressed also more positive thoughts that overturn this impression. These views are represented in the workshop participants desires to ask for more services. Simultaneously the workshop participants expressed the will to have the smart homes services and functions that serve their everyday life and *improve the life*.

This is described by Stavros:

"Yes, and if the system irrigates the flowers! It is important invention. Because, when someone leaves for holidays, says "Who will water my flowers?"

Also, there is another belief, expressed by workshop participants. They emphasised that the use of technology it's up to the residents. They decide if they will use the technology in a way to solve problems or make their life more complicated.

The *use of out-of-scenario knowledge* about the technology is presented in the statement of Nikos:

"You use it in a good way? You get benefits. You use it in a bad way? You got problems. It is upon the use of technology."

The considerations of the workshop participants and their awareness of the smart homes functions and services were expressed in this theme. The views of the participants present a distinction between the acceptance of the services and concerns about issues such as the vulnerability of the system and its limitations.

6. Discussion

In this chapter the findings of the research are discussed. In this procedure I used the theoretical framework and the review of literature that was deployed in this thesis.

The aim of the study was to investigate Internet of Things as an emergent research area with a specific focus on the design and implementation of smart homes. This research has a focus on humans interactions in smart homes. Three research questions were formulated to investigate the how, what and why of the phenomena smart homes. This deliberation of the empirical findings provides "a discussion of the degree to which the research has answered the questions" (Hart, 2005, p. 100). The outcomes help also to compare the elements of the research with the perspectives of the Theoretical framework and the Literature review as they have been presented. The research was conducted through a qualitative research approach and used the theories of phenomenology, postphenomenology and phenomenologically-situated paradigm of HCI as the theoretical framework that deals with the relation of humans and technology and the shaped *lifeworld* (Dusek, 2006, p. 71). Also, the methods of the research based upon interviews, workshops and observations that support the common aim to understand the internal and implied causes of the phenomenon in the lived experience of the humans (Lichtman, 2013).

The first research question was: *In which way are designers and prospective residents involved in design of smart homes?*

The answer of this research question is based on the outcomes represented in the themes of *Design of Smart Homes* and *Home Control*. The designers are involved in the development of the requirement specification. They collaborate with the prospect residents to decide the type of the infrastructure that would be useful to install. Also, designers use interviews or surveys to identify prospective residents' desires to propose a set of equipment and functions that apply to the residents' wishes. Further, the designers take into account the equipment, the interoperability of the devices and the environmental conditions. The subcategories *home automation* and *distance control* refers to the functions and services of a smart home like the automated close of the windows that can be performed by distance, through the internet, and the use of smartphones, tablets, etc. The use of infrastructure (Jahn et al., 2010) refers not only for the specific requirements of a smart home, but also and further in Internet of Things operation (Atzori, Iera and Morabito, 2010). The infrastructure is constituted by many types of devices like sensors, actuators, dimmers and controllers and may be materialized through structured cabling or wireless (Zhang and Zhu, 2011; Li et al., 2011; Welbourne et al., 2009; Qin and Gu, 2011). The implementation of a smart home follows the approach of the designers of what it should be included in a smart home. The subcategories *easy living* and *safety and security* that constituted the

theme Home Control illustrated also how the designers were involved in the design of Home Control. Designers have their own vision regarding the services and functions of smart homes. They repeated the concepts safety, convenience, security, etc. This perspective provides a reasoning that leads to the thought of an *embedded morality* (Verbeek, 2006) into the services that the designers install. Also, it implies an intentionality of human consciousness and actions (Introna and Ilharco, 2004). On the other hand the workshop participants request a house that facilitates their everyday life and is represented in the term of *easy living*. In this direction they seek a range of functions that includes, lighting control, energy saving, safety, security and default functions (Wu and Fu, 2012; De Silva, Morikawa and Petra, 2012; Chan et al., 2009).

Design of Smart Homes and Home Control are constituted also on the subcategories *clients needs* and *clients information*. These subcategories provide the experiences of the humans in collaboration tasks. The designers provide to their clients information about the functions and the services of smart homes and on the other hand the workshop participants express their views and desires that form the type of infrastructure in the home. In addition, they discuss with the prospect residents the aim and the scope of the installation. There are different views between the specific needs and preferences of workshop participants and the designers' perceptions of the projects on the level of settings of the system and the installed devices. The *customization* is a main element on the design of a smart home and is based to the residents' experiences which are shaped by socio-cultural factors (Ihde, 2009, p. 18-19). The clients and prospect residents act as human entities that create their needs according to the social environment that affects their attitude, and also anticipate the use of the technological artifacts as *transcendental* domain, the sense of the ongoing experience that helps them to perceive the world as it is (Introna and Ilharco, 2004). The workshops participants refer to their experiences with the technological objects like smartphones, cameras, automations, etc. that reveal an already reminder experience about the things that determines an intentional and conscious representation of the objects (Flores Morador, 2011).

Furthermore, the subcategories *distance control*, the *cross-platform interface* and the *free of commitment of the physical presence* are presented in designers' views and workshop participants are aware of the capabilities of the system and the services (Vermesan et al., 2009; Chan et al., 2009). Smart homes provide the residents the capability to control the home by distance and also to give access to the family members when they are at work. Also, the residents can simulate a presence at home with a turn on/off of lights or the opening and closing of shutters. It is interesting that the workshop participants accepted the featured services on the demonstrated videos and build up their own "cases" according to the services and functions. These functions are facilitated with the use of smartphones, tablets and of course through the IoT that permits the communication among things and humans (Azzara et al., 2013). In addition, the subcategory *notifications by the system* also emerges. The system uses *notifications* about the status of the home devices and conditions and

provide to the residents information that help them to take decisions (Robertson and Wagner, 2015). This procedure has similarities with the concept of Computer Supported Cooperative Work (CSCW). Any device or part of the system collaborates with the others in order to achieve a common and final goal (Schmidt and Bannon, 2011, p. 5).

Also, the factors of *safe and security* are an issue that constituted the importance of smart homes. The interviewees and the workshops participants were paid attention to the safety and the security in smart homes. This is explained by the nature of humans that want to feel safe and to live in a comfort and healthy environment that could make them feel happier (Chen, 2012). The *safe and security* refers to the functions of controlling the appliances to avoid damage, like a fire by an open oven, and furthermore the secure close of windows and doors with the activation of alarm system to prevent burglaries. The workshop participants and interviewees emphasized security and safety functions like to secure that the windows and the doors are closed. They consider these as core elements included in the smart home. Also, the surveillance performed by a set of cameras and sensors is an added layer of security in smart homes. The subcategories *requirement specification, clients needs, perspective of the Internet of Things, clients information, easy living, cross-platform interface, simple interface, home automation, notifications by the system, free of commitment of the physical presence, distance control, safety and security, customization* of the themes that were discussed produce a complex frame of conclusions that answer to the first question. There are different views expressed by the designers and workshop participants. The objections of the designers are intersected with the workshop participants' needs and desires and converge in order to produce a balance that creates the smart home environment.

The second question was: *What kind of services and human-system interaction do designers include in the design of smart homes?*

The answer to this question is coming from the identified theme *Interaction with the system* and also with the use of some elements from the themes *Home Control* and *Design of Smart Homes*. The subcategories *clients needs, customization, simple interface* and *technological mediation* are interdependent components that draw the boundaries of human-system interaction in smart homes. There is a communication between designers and prospect residents about the functions, the services and the infrastructure of smart homes. The workshop participants express desires that extent from heating control to watering the flowers. The human-system interaction is a subject composed by many concepts of Human Computer Interaction such as the understanding of salient properties of interaction, the human factor, the space, the questions and the answers that arise from the properties of interaction and the interpretation of these questions (Harrison, Tatar and Sengers, 2007). The *perspective of the Internet of Things* is another subcategory that is included in empirical findings.

The installed infrastructure of smart homes like sensors, actuators, cameras, thermostats or dimmers are interconnected and communicate through the central server of the home. This communication is extended to the IoT through the web and the middleware level of the IoT that permits the interoperability of software and smart objects. In this case of human-system interaction in smart homes, the IoT is involved as a platform that supports the infrastructure and as mean of the interaction (Guo et. al., 2012; Vermesan et al., 2009). The prospect residents use devices and functions that permit them to do settings, to intervene in the home status or to communicate with the other residents, remote or local. These features of interaction have a great analogy with the concept of *technological mediation* (Verbeek, 2006) and the layer that technology creates between humans and their perception of the world. In this procedure people use the artifacts (devices, functions) as a part of their bodies, they embody the artifacts, so the technological objects are not the end of human perception and action, but withdraw their attention, making vivid specific experiences and practices (Verbeek, 2006; Ihde, 2007). This transformation of the reality changes the nature of the things that the human anticipates as objects with alternative shape and properties. The capability of the distance visual control of the home is the result of the *metaphor* of the home plan and settings in a display that allows the user to control the home by distance.

Therefore, this concept of *metaphor* of home spaces and settings has implicated with many issues of the third paradigm of human-computer interaction like dynamic use context, socially situated, indirect and multiple goals, non-task oriented computing, handling emotions and the problem of *phronesis* (Harrison, Tatar and Sengers, 2007). The actions and the understanding of the residents about the objects and systems are socially situated. They are affected by the social conditions, for example the workplace or the home environment. These environments form the context of use and help the construction of meaning of services and devices. For example, the presence of cameras and sensors may affect the decisions the residents make and their attitude towards others during a meeting. The workshop participants express their desires and concerns about the services and functions of smart home regarding to what is good or bad for them. This is summarized in the issue of *phronesis,* that constitutes the "wisdom of ethical matters" (Harrison, Tatar and Sengers, 2007, p. 6). The resident should take a decision in an environment that surrounds him and influence his behavior. The depiction of the home in the screen creates a meaning that can be perceived, in this context with a different way about the things, such as the meaning of the things in any particular time depends on the remainder, the already *sense* that we have for the objects (Introna and Ilharco, 2004). Furthermore, the use of system capabilities produces another belief that the residents do not have to remember about the functions of the home or that this remembering process is supported by the system. Workshop participants wish to include a sense of "care" to the smart home system providing to the residents the opportunity to delegate some tasks to the technology. The service of monitoring the home by the

interconnected devices, that interchange data to each other, shape a system that forces the residents to delegate the remembering to the system. The *loss of remembering* (Maylor, 2014; Afyounian, 2014) is another issue that arises in the interaction of humans with the system. The workshop participants see the smart homes services as a chance to delegate tasks to the system. They believe, that in a smart home -as a computing environment- the resident does not need to consider remembering whether the oven or the window is closed, the smart home will do it for them. The system interact with the residents through a set of *notifications* and information that supports the life of the residents, as they believe, producing a sense of safety and security. But, at the same time, all the procedures of this interaction examine a grade of power that is running within the orbit of mechanisms of technology and the human action as Foucault suggests (Willcocks, 2004). The normalization of the behavior is a subject that arises through the use of technology. Humans use a predetermined system that sets the boundary of their attitude. The individuals are becoming part of a net that operates upon an endless structure of devices, procedures, legislations and limitations. The concept of CSCW confirms that (Schmidt and Bannon, 2011). In this context the residents and the devices act as components of "cooperation" among the other users of home and the functions of the smart home. It is useful to consider that smart homes are part of IoT realm. In addition, another subcategory is the *change of habits* of smart homes residents. The facilities that a smart homes offers, like the distance control of the appliances, doors and windows or the programming of a sequence of functions like the heating of water in the bath may provide the residents the opportunity to change habits. To take a bath at the morning or to open and close the door when the kids are returning from school without the need of physical presence are some of the everyday activities that may change. Technology transforms the reality of humans and beyond that creates the conditions to *change habits*. The workshop participants expressed the view that they may need to change their everyday habits if the system provides the facilities. This concept that technology helps or forces the people to change habits is obvious in the vision of the Social IoT and in the offered services (Guo et. al., 2012). The users and the objects interchange information, take decisions and create content in the web (Michahelles, Karpischek and Schmidt, 2010). But, this flux of information and the sense of opportunity to act, turn into a procedure to change habits and behavior according to *lifeworld* that they perceive (Dusek, 2006; Guo et. al., 2012). Following this reasoning, another implication in the human-system interaction arises for the prospect residents, that is the fear of obligation to monitor the system (Chan et al., 2009; Zheng and Feng, 2012; Fahim et al., 2012). Although, the system is designed to provide to the users the ability to monitor the house, the prospect residents have the sense that they ought to supervise the system and its functions.

The *ethical issues* are another factor that affects the human-system interaction. The prospect residents concern about their privacy and the protection of their personal data. The workshops participants discussed the security of the wireless infrastructure

and believe that there will be some danger due to the web connection. On the other hand the designers assure them that they have all the requirement protection layers in the software that decrease the possibility of a breach. The system is considered such a mean that could be used by a stranger to intervene in their private life (Weber, 2009; Atzori, Iera and Morabito, 2010; Wu and Fu, 2012). According to this the prospect residents adjust their behavior and use the options of the system to disable e.g. the cameras in order to protect their personal data and private life. This behavior guides, in a wider context of Social IoT, to a division between the open persona and paranoid persona (Michahelles, Karpischek and Schmidt, 2010). It is real that users have adopted that their privacy is a subject of conflict between the commercial use of the data and the democratic rights of individuals' and more forward to the ethics. From the other hand a defensive and refusing stance to share and communicate data constitutes the profile of a paranoid persona. Furthermore, the use of *scenarios* in the human-system interaction has a particular value in the prospect residents' view of smart home environment. The separation of the family members in a distinctive place and time context into the same home is a subject that the dedicated scenarios of smart homes realize. The specific preferences of each family member are becoming a corpus of services that the system offers and both designers and residents consider about (Du and Wang, 2012).

Although, the workshop participants acknowledge some limitations in the use of scenarios, they have a positive stance about the use of smart homes. They expressed their views, feelings and experiences related to the technology, smart homes and ethics. The human-system interaction in smart homes is an issue that can be understood with the concepts included in the theoretical framework, like the *technological mediation, embodiment, lifeworld* and the infrastructure that materializes the IoT realm and the smart homes.

The third question was: *What visions and ideas do prospective residents express about life in smart homes?*

This question investigates the way the prospect residents of a smart home imagine the life in this computing environment. The lack of the contact with existing residents of smart homes was replaced with two workshops where the participants had the opportunity to watch three short videos that demonstrated functions and services included smart homes. The participants expressed their feelings, desires and concerns about the life in smart homes. These feelings and desires were *Ethical Issues, Awareness and Considerations* and *Interaction with the System*. Ethical Issues refer to the considerations that arise from the use of cameras in smart homes, the sense that the system intervenes to the humans' relations, the protection of privacy, the fear of breach of the system and the view that the system threats the humans' activities and tends to replace them. Awareness and Considerations included the views of the workshop participants about the smart homes services and limitations, the

vulnerability of the system, the importance of human factor in the security of the system and their previous experiences that form their standpoint about smart homes. Furthermore, the interaction with the System refers to the concepts of technological mediation, the loss of remembering and the outcomes of the empirical findings analysis that smart homes provide the opportunity or force the residents to change habits. Also, the support of living is another element that constituted the theme Interaction with the System. The subcategories of these themes correspond to the participants' point of view for the automations of a smart home, the considerations about the morality of the design and the implications in their personal and social relations.

The *protection of privacy* and that the *companies protect the system* are two interdependent subcategories. The main issue that the participants take into account is the *protection of privacy* that is combined with the assumption that companies have the capability to collect and store a huge amount of data. They expressed worries about the possibility of data collecting that might have consequences to their privacy (Weber, 2009; Guo et. al., 2012). Although, the designers claimed that they do *not collect data* and there is also the sense of the workshop participants that *companies protect the system*, they still expressed some doubts about the level of the protection and the human factor that could intervene in the system and consequently in the residents' life (Gubbi et al., 2013). Through the interviews and the workshops it is described a distinctive difference between the designers and the workshop participants beliefs. This disagreement interprets the different meaning of the things that is dominated from the factors such as the *embodiment* of the materials/things, the technique of use and the cultural-historical role of the environment where an *experience* is generated (Ihde, 2009, p. 18-19). The designers expressed the view that residents can control their home by smartphones, tablets and PCs that they bring with them or use in their office. Users embody these devices and the devices withdraw from their attention (Ihde, 2009). Furthermore, the *sense of loss the personal contact* and *system intervention to the humans' relations* are critical in their perception of smart home environment. Here, is emerging the subcategory *technological mediation* of the theme Interaction with the System. The workshop participants feel that the functions of the system intervene in their relationships with others and experience a risk to lose the contact with family members and friends. The workshops participants have to experience the living environment of a home through the complex contribution of a technological grid that *mediates* between the world and humans and creates the *lifeworld* within the humans act. This illustrates how a smart home has consequences on humans' life and the implications of this relationship to the humans understanding of the reality (Dusek, 2006; Verbeek, 2006).

Therefore, the workshop participants posed a positive stance when the workshop conversation focused on the functions of the smart homes that are absent in a traditional house. For example, the self-configured heating system depending on the environmental conditions and the automated turn on of the lights when they wake up

are some functions that included in smart homes. So, they characterised these functions as useful and also expressed their acceptance of services with a declared thought that they could *improve the life*. The workshop participants want to make their life easier and in this direction they request services to facilitate their everyday life. Their wishes are similar to a virtual "housekeeper" (Du and Wang, 2012) that take care of the house.

When the participants watched the capabilities of the system and dealt with the scenario of the workshop that was built on everyday life in a home. They were persuaded that the life in a smart home is facilitated much more in terms of this infrastructure. It is impressive that the participants use their experiences of the family life and the roles of the family members in order to deal with the concepts of technology. This reveals that people need to feel special, stay healthy, avoid unhappiness and save time (Chen, 2012), even with the support of technology that at the same time imagines as a threat, which can substitute the human actions and deskill people. So, the workshop participants deny delegating to the system some actions that attach to humans' relations. They prefer to prepare a coffee for their visitors or to go to their child bedroom to close the lights or windows. But, they accept the operation of the heating control and alarm activation by distance when the children are alone at home. It seems that technology, in this case, performs some properties that allow humans to make a distinction regarding their ability to achieve the operations that are intended to serve. When a tool or machine executes a work better than a human or than another tool is attributed as "whole" technology artefact. From the other side, when this performance is inadequate then the artefact constitutes a "broken" technology (Flores Morador, 2009, p. 17-18).

Also, this distinction among the functions and services of smart homes expresses the will of the individuals to set the boundaries of their *lifeworld*. Technology is not neutral and humans use it to interact with artifacts. Technology has an *embedded morality* by its designer's intentions that affect the behaviour and the experiences of users (Verbeek, 2006) and forms the term of persuasive technologies. As extracted from all the previous discussed elements the view of the workshop participants about life in a smart home hasn't formed a definitely positive or negative stance, but is constituted from a complex grid of considerations about ethical issues, desires for services, acceptance of use for some functions and refusal for some others. This concludes to a positive thought, that smart homes improve the life of the residents, and the significant perception that the use of technology depends on the residents' personal needs and experiences.

7. Conclusion

This section offers conclusive remarks, the contribution of the study, reflections on the research process, and possibilities for further research.

The aim of this research was to investigate the Internet of Things with a specific focus on the design and implementation of smart homes. This research has a focus on human interactions in smart homes. The empirical findings formed a context that shapes the relations between the humans and the computing environment of smart homes as part of IoT realm.

The contribution of this study to the research field is a human-centred perspective of human interactions with the smart homes system as a practice of IoT. In my research I focused on experiences of the participants and their visions about smart homes design and human interactions. The use of workshops as a method of Participatory Design indicates the co-design process as a solution to design technology that meets the needs of people in an effective and attractive way. The theories of Phenomenology and Postphenomenology helped to reveal and interpret the importance of human experiences and the relation of the technology and humans that form the *lifeworld*. In addition, I paid attention to the concepts of *technological mediation*, *embodiment* and *embedded morality* that draw the boundaries of the design and use of the artifacts and the human perception of the reality through the use of technology.

The research has as aim to contribute on the field of smart homes and human interactions and to help both designers and users to understand the interdependence of their role. As a researcher on the human-system interaction in smart homes I believe that I have placed a piece of knowledge on the existing bibliography.

This research constitutes a subject for discussion for its outcomes and contribution. A further investigation about the human interaction in smart homes and human-centred IoT could extent this research impact and could offer an advanced level of knowledge.

7.1 Reflections on the Research Process and Outcomes

As a researcher I had the opportunity to engage in a very interesting and difficult process to construct a framework of a qualitative research approach for this study and parallel the responsibility to collaborate with the participants and to respect their personalities and experiences according to the ethics. The same attitude I followed for all period of the research both to the authors of the literature that I used and to the readers of the present thesis.

In the process of research the workshops were a challenge for me, but at the same time, I believe they brought me closer to participatory design (Crang and Cook, 2007; Bannon and Ehn, 2012; Mörtberg et. al., 2010). I think the disadvantage of not managing to conduct interviews with already users of smart homes turned to an

advantage, because I had the chance to understand the perception of the phenomenon by the possible prospect residents and to observe their reactions in the workshop scenario and in the demonstrated videos. Also, the direct comparison of the life in a traditional home was clear and fruitful for the research. Interviewees and workshop participants expressed their experiences about smart homes and offered their views, assumptions and visions.

The outcomes of the research answered the three research questions. The research answered "*In which way are designers and prospective residents involved in design smart homes?",* of the first research question. The contribution of the designers is realized through the communication with the prospect residents. They inform the clients about the offered services and functions, the types of infrastructure and propose solutions. Also, designers have their own views about the aim and the scope of an installation and these are revealed in their visions of smart homes implementation. On the other hand, the prospect residents contribute in the process of design of smart home with their desires about the services and their views and considerations regarding the use and implications of smart homes to their life. The "*What kind of services and human-system interaction do designers include in the design of smart homes?",* of the second research question. The designers create a system where the residents feel safety and security. The services that these systems offer like the automations and distance control support the living of the residents and make their life easier. The residents interact with the smart home through simple interfaces that are provided by a wide range of devices (e.g. smartphones, tablets, embedded displays), a set of notifications and visual depictions of the home plan and customization of settings. In addition, the system facilities provide the residents the opportunity to delegate tasks to the system and change habits. The remembering of the appliances operation or the functions such as the secure close of windows is a task that the system performs. Also, the system creates a technological mediation between the humans and the living environment that may affect their perception of the devices and services. The "*What visions and ideas do prospective residents express about life in smart homes?",* of the third research question. In the workshops it is expressed an overall positive stance about the life in smart homes. There is the view that smart homes improve life. Although, were expressed concerns about the issues of privacy, the protection of the system by breach and the possibility that companies may collect data about the habits of use and the residents private life. Ethical considerations about the human personality and activities that may be threatened by the use of the system were also formulated.

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Appendix A

Informed consent form

Researcher: Paraskevas Kamatsos

Institution: Linnaeus University

Dear participant,

This research is conducted as a Master Thesis in the M.Sc. in Information Systems and will be presented on the department's examiners and students. Also, the thesis will be published in the Open Access institutional repository of Linnaeus University.

The subject is the *Smart Homes: human interactions and IoT*. The aim of the study is to investigate the ways and reasons that the humans interact with the computerized environment of smart homes.

The method of the study is the qualitative research with the personal interviews of individuals that live in smart homes.

The participation on the study is optional and you can withdraw at any time. Your personal data will be kept confidential.

I will respect your personality, opinions, thoughts and beliefs during the research procedure.

Researcher Participant

Appendix B

Interview questions

1) Please could you tell me about your educational background? And also about your working life experiences?
2) Now do you work in a smart home project? Could you describe the project? When and why did the project started? Does your company started the project or is a consultant? Do you have previous experiences with this kind of projects? When were you engaged in this project? And why? What was your "vision" for the project? Who do you believe that will be living in this kind of smart home?
3) This kind of project has an overall objective, what is the objective in this project? Also, in this overall objective, there are some specific requirements that are called requirements specification, are you familiar with that? Have you been involved in the development of it? Could you please describe the specification?
4) Have you taken into account the required specifications?
5) There was a vision about who is going to live in these apartments/houses, did this vision also include more specific persons e.g. aged people, impaired persons, etc.?
6) A smart home includes various services. Could you please give me some examples of the services that are offered? How do these services support the residents? Any hindrances?
7) What was the technological equipment that you had used on the project?
8) Did you involve the residents on the design? How did they become involved?
9) Did the residents have some specific demands for the smart home implementation? Could you exemplify their demands?
10) Now you have the opportunity to install various kinds of technologies. What kind of technologies will be installed, could you give me some examples? In what way do these have an impact on the residents' day-to-day life? Support? Obstacles?
11) Do you believe that the smart home technologies may support the day-to-day life of the residents? How the residents control the smart home equipment? Do they need to have specific skills to navigate in their homes?
12) What is the concept of the project? Do you consider the project as part of Internet of Things? What are the negative or positive elements in this perspective?
13) By implementing these technologies on the smart home, some ethical issues may arise like the reveal of the residents habits through the use of the

equipment, for example the TV, lights, washing machine, etc. Do you consider this kind of ethical issues?

14) How did you handle them?
15) Do you keep data about the use and the function of the smart home?
16) What kind of data they are? They are data regarding the behavior of residents or the equipment?

Appendix C

Workshop scenario

[The scenario describes happenings of a common day in a smart home]

7:00 We wake up

The lights on...

The shutters and the windows open...

Radio / TV play in the news...

The coffee is ready...

7:30 We are leaving

The doors close...

Lights off...

Alarm activates...

10:00 [On the job] We check from the smartphone/ laptop

Kitchen....

Outdoors....

Entrance camera....

Phone calls....

11:00 [On the job] Visit of plumber to repair the damage

Notice on your smartphone that the doorbell is ringing...

We check the camera...

Disable alarm...

We talk to him...

13:00 [On the job] The child returns from school

We check entrance camera...

We open the door....

Disable the internal sensors of the alarm and leave the outside...

Lock the door...

16:00 We return to home

The garage opens when we get closer and it recognizes us...

We go home...

The heating is adjusted to the weather of the day...

The TV has opened up to the programmed channel...

17:00 Visit of friends

We check entrance camera...

We open the door...

18:00 Sun has been lowered

The blinds open and lights dimming…

Check the temperature in the child's bedroom...

The heating is adjusted...

21:00 Friends are leaving

We open the entrance door...

Disable the external alarm sensors...

We close the entrance door...

22:30 We are going to sleep

We set the house in sleep mode…

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